

The Short-run Effects of Opening Mobile In-App Payment Systems: Evidence from South Korea

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Abstract

Mobile app platforms are highly concentrated – Apple and Google each distribute over 75% of the total number of apps installed on the relevant devices. These platforms generally require developers to use a built-in system for processing payments with a commission rate of 30% for both the initial purchase of an app and any subsequent in-app purchases. In September 2021, South Korea became the first country to ban this lock-in; purchases made in the country may be conducted through any billing system developers wish. We analyze the impact of this policy change (and Apple’s subsequent adaptation) on demand for apps and revenue using difference-in-differences techniques and data on apps offered through Apple’s App Store from a leading app analytics firm from January 2018 to December 2023. We find no evidence that the policy change generated substantive changes in South Korea’s app marketplace.

JEL Codes: K21; L12; L22; L41; L89

Keywords: Platform, In-app Billing System, Tying, Taxation, Industrial Organization

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1 Introduction

Many digital platform markets are highly concentrated, dominated by a few players who have built a large share through network externalities (Rochet and Tirole, 2003). These players often employ ‘tying’ strategies to extend their dominance in one market to other markets (Carlton and Waldman, 2002). By foreclosing rivals, tying can lead to the monopolization of the market for the tied good (Whinston, 1990). For example, Microsoft once tied its Internet Explorer browser with its Windows operating system to hamper the growth of competing browsers. The lack of competition from the exclusionary conduct associated with tying raises several anti-competitive concerns. The price of tied products is likely to be above the competitive equilibrium price and the level of product variety is likely to fall, both of which contribute to a decrease in consumer welfare (Whinston, 1990). It can also negatively affect the level of innovation in the tied good market as potential entrants choose not to invest, which in turn further strengthens an incumbent’s dominant market position (Choi and Stefanadis, 2001).

Our study focuses on the market for mobile apps, which features substantial tying. Apple and Google, the leading firms in the market for mobile operating systems, each operate a distribution platform (Apple’s App Store and Google’s Play Store) that has near-monopoly power over the sale and distribution of apps to the relevant end-users. These firms have used the power implicit in their access to a large base of mobile users to dictate contract terms to app developers. Among these terms is a requirement for developers to exclusively use the platform’s payment system to process in-app purchases. On both platforms, 30% of every purchase is remitted to the platform owner as a commission. These contracts have been subject to public, legislative, and judicial scrutiny in many countries in recent years as developers of large and popular apps have complained that the current system is unfair and limits development effort (see, e.g. Allyn, 2021). The most successful effort, from the perspective of app developers, is a 2021 South Korea law that banned app store operators from enforcing exclusivity requirements for in-app payment processing.

We study the effects of the law and the subsequent responses from Apple on the app market. The law came into effect on September 14, 2021, though Apple did not change its policy until June 30, 2022. Developers of App Store-distributed apps may offer South Korean users an in-app billing system that differs from the App Store’s built-in billing system if they complete required forms and

agree on additional terms, including a still-substantial 26% commission.¹

We develop hypotheses based on the claims of app developers: since the commission payment acts as a tax, a reduction in the tax may lead developers to improve the quality of their apps (Hwang and Kim, 2022). This in turn may increase the number of users with the apps installed and increase the income of developers from in-app purchases. We validate this possibility theoretically with a simple model of app development and sales. On the other hand, as argued by platform operators, users may be distrustful of alternative payment platforms and any benefit to developers may be more than offset by a shift in demand (Schreft, 2007; Eriksson et al., 2021).

We test for changes in app-level outcomes using difference-in-differences and triple difference-in-differences techniques. Our data consists of install and revenue data from Appfigures, a leading industry analytics provider. We focus on apps with substantial activity in South Korea around the time of the policy change. Ultimately, while our point estimates suggest that a small increase in downloads was more than offset by a decrease in the number of in-app purchases to result in slightly decreased revenue, our estimates are noisy and we therefore conclude that the policy change did not generate any substantial changes in outcomes. We attribute this primarily to the small size of South Korea relative to the global marketplace for apps and the way in which Apple responded to the policy change.

Our work contributes to the broader literature exploring the theoretical and empirical implications of tying strategies. Within this literature, most of the work has focused on examining the effects of tying (Carlton and Waldman, 2002; Amelio and Jullien, 2012; Derdenger, 2014; Choi and Jeon, 2021). In contrast, studies that analyze the effects of attempted remedies and/or regulations that seek to unbundle once-tied products are relatively scarce, in part because such regulations are themselves scarce. Gans (2011) shows that under certain conditions, mandatory untying policies may be effective at increasing social welfare. Decarolis and Li (2023) examine the EU’s attempts to untie mobile search in its jurisdiction and hypothesize that the success of any untying policy depends on the existence of “qualified challengers who can compete [on] quality.” We contribute to this literature by providing an empirical analysis of the first app platform untying policy enacted globally.

Our work also contributes to the literature on mobile app platforms. Since there are multiple platforms available, a primary topic of interest has been multihoming, which has been shown to

¹Google had earlier changed its policies in a similar manner.

have ambiguous effects on prices and platform fees both in theory and in different empirical contexts (Landsman and Stremersch, 2011; Belleflamme and Peitz, 2019; Teh et al., 2023). In the mobile app context, Hyrynsalmi et al. (2016) show that while multihoming is rare when considered against the broad population of apps, multihoming is much more common among the set of ‘superstar’ developers and apps; Ershov (2024) shows that platform congestion is a significant externality in this context. Gans (2012) provides a theoretical basis for the commission rates charged by platforms by demonstrating that charging for platform access (rather than taxing platform revenue) leads to an unravelling problem. Miao (2022) shows that the commission rates charged by platforms may *increase* social welfare by dampening the revenue-shifting incentive faced by firms selling ancillary goods (such as in-app purchases).

Quantitative empirical work on in-app payments is limited due to the limited set of reforms or instruments useful for identification, though some qualitative analyses exist (see e.g. Ravoniarison and Benito, 2019; Hamari et al., 2020; Salehudin and Alpert, 2022). One relevant qualitative analysis is that of Hwang and Kim (2022), who study the Korea reform and find that network effects and government guarantees against retaliation from Apple or Google drive decisions to switch to alternative billing systems or app distribution platforms. On the empirical side, Enache et al. (2023) use an exogenous adjustment to Apple’s policies in 2021 along with data from five “freemium” mobile app games in European markets to estimate own-price elasticities of conversion (i.e. the rate at which price changes generate changes in the propensity of users to convert from free to paid status) in the -1 to -4 range.

A growing body of literature provides a theoretical framework that examines the role of fees for platforms, although not many studies have directly and empirically focused on the app store market. For example, Gomes and Mantovani (2024) build a theoretical framework to derive the optimal cap on commissions, concluding that the cap should equal the amount of informational and convenience benefits provided by the platforms. De Cornière et al. (2023) develop a model demonstrating that the third-degree price discrimination in two-sided markets can improve welfare compared to uniform pricing. Similarly, Wang and Wright (2018) compare two different fee structures, fixed per-transaction fees versus ad valorem fees, and find that ad valorem fees generally increases the welfare. We introduce a simple model of profit-maximizing app development to motivate our analyses, but take as given the commission rates charged by platform owners.

We proceed in Section 2 with a brief background on the mobile app market and the in-app billing

policies of the major platforms. We introduce our data in Section 3 and provide some descriptive evidence. We detail our empirical approach in Section 4 and discuss our results in Section 5. We conclude in Section 6.

2 Background

2.1 App Stores

App stores are digital platforms for selling and distributing mobile applications (apps). Developers publish their apps on these platforms while consumers use the platforms to discover, purchase, and access apps. Developers earn revenue through three methods. First, developers may charge an up-front price to download the app. Second, developers may embed advertisements into the app. Third, developers may lock certain features behind in-app purchases. These methods may be combined. For example, a strategy game app may be offered to download for free, but come with advertisements displayed between each round or before each turn. Users may pay a fee to remove such advertisements. The app may offer additional features in exchange for fees such as in-game performance bonuses (e.g. increases to the rate at which the user progresses through the game) or changes to the appearance of game features (e.g. ‘cosmetic’ changes to the characters in the game that do not affect game outcomes).

App stores offer users (among other features) limited security guarantees, a centralized record of purchases, and convenient re-installation of apps on new devices (e.g. a user changing mobile devices which both use the Google Play Store may reinstall their applications on their new device with minimal intervention). These platforms exclusively offer developers access to large user bases (i.e. developers who wish to offer their apps to Android users must do so through the Google Play Store) and baseline functionality related to online communication, payment systems, and cloud-based storage (of e.g. saved documents or play sessions). App stores generate revenue primarily by charging a commission on both purchases of apps with up-front prices and in-app purchases.² Historically, the ‘tax rate’ charged by both Google and Apple has been set at 30%. In other words, when a user spent \$10 on either an up-front or in-app purchase, the developer received \$7 and the platform received \$3. In early 2021, both Apple and Google reduced the tax rate to 15% for the first \$1 million of revenue earned by a developer each year.

²Google also offers a monthly subscription service that includes access to a rotating library of paid apps.

Other app stores exist. For example, three major South Korean telecommunications companies maintain the ONE store platform which is installed on all mobile devices sold in South Korea that are serviced by the relevant wireless networks. ONE store offers developers a lower commission rate of 20% when using the ONE store payment platform and a 5% service fee when using other platforms.

2.2 The South Korean Law and Responses

Responding to antitrust concerns, the South Korean National Assembly banned the operators of app store platforms from imposing payment methods on app developers starting September 14, 2021. This policy change made South Korea the first country to compel the opening of in-app billing systems by law.

The law required platform operators to submit detailed plans about how they would change their policies to comply. Google implemented changes in December 2021 that allowed developers to offer users in South Korea an alternative in-app billing system. To do so, developers must certify compliance with industry standards for security and fraud prevention, report all transactions from users in South Korea on a monthly basis, and pay a per-transaction service charge equal to the non-South-Korea market rate minus 4%. In other words, while developers may offer users an alternative payment system, most payments are still taxed at 26%. Apple followed suit in June 2022, mirroring Google's changes (including the 26% commission rate).

Regulators in other countries are considering similar actions. The Netherlands' antitrust authority has ordered Apple to allow alternative payment methods for dating apps. In the United States, the Open App Markets Act is a bipartisan effort to untie payments and implement additional restrictions on app platform providers; the legislation was drafted in part in response to a lawsuit from a prominent app developer challenging Apple's in-app payment restrictions.

3 Data and Descriptive Evidence

To study the short-run effect of South Korea's legislation, and the corresponding policy responses of Google and Apple, we obtain panel data on app downloads and revenues from Appfigures. Appfigures is a firm that provides developers tracking and analytics services. Their data has been used by researchers in a number of fields to study app marketplaces (see e.g. Coughlin et al., 2016; Su et al.,

2021; Pybus and Coté, 2024). We obtain Appfigures’ dataset of app installations and revenue net-of-commission fees for the 174 apps that matched the following criteria: (a) available on Apple App Store and Google Play Store, (b) monetized solely through in-app purchases (i.e. free to download), and (c) were “active” in the sense of positive revenue and downloads in the South Korean market whenever they were available on the App Store. We obtain this data at the country-month level from January 2018 to December 2023 for South Korea, the United States, Japan, Germany, and France. We transform net revenues into gross revenues.

It is important to note up front that these data consist of Appfigures’ estimates of the relevant metrics. Appfigures forms these estimates by training prediction models on private data reported by developers who opt-in to its data collection program.³ We proceed under the assumption that the Appfigures data are noisy measures of the true install and revenue data that have neutral bias relative to the reform. By ‘neutral bias’ we mean that to the extent that the Appfigures estimates are biased measures of the true data, that bias is not affected by the reform. Thus, comparisons of app performance pre- and post-reform can identify at least the direction of any effects.

We chose our inclusion criteria as the apps that fit our criteria are the most likely to be impacted by the legislation, given their primary revenue source is in-app billing and their exposure to the South Korean market. Any effects of the policy change of Apple’s App Store in response to the legislation should manifest within this group of apps that rely heavily on in-app purchase models and have a substantial presence in the regulated region.

We chose the comparison countries as their per-capital spending on mobile apps is similar to that in South Korea. According to Statista, through the first three quarters of 2021, Japan had the highest mobile spending per capital with \$149 US. South Korea was the second with \$95 US, followed by the United States with \$90 US. Germany and France were 7th and 8th, respectively.⁴ Although similar legal movements were put forth by developers against the two major app distribution platforms across several countries during the sample period, we observe no significant changes to the commission rate structure imposed by the Apple App Store until the end of 2023. This allows us to use the app performance data from these unaffected countries as a control group for our analysis.

³In other words, Appfigures observes publicly-available data from Google and Apple for all apps, and private data from developers that opt-in. Appfigures trains prediction models on the set of apps for which private data are available, and produces estimates of the private data for all apps. It is these estimates that we use as the outcome measures in our analysis. Using the industry standard Mean Absolute Percentage Error (MAPE), their reported error rate falls in the range of 5% at best to 25% at worst. See details <https://appfigures.com/support/kb/680/how-to-compare-your-downloads-and-revenue-with-estimates-in-competitor-intelligence>.

⁴<https://www.statista.com/statistics/1275323/top-countries-mobile-per-capita-spending-worldwide/>

We specifically analyze these apps’ performance on the Apple App Store for two reasons. First, Apple’s in-app purchase policies and commission rate policies have remained stable over the period examined, unlike Google Play. This allows for a cleaner comparison of app performance pre- and post-treatment. Second, this analysis controls for substitution effects across app stores, as iOS users are inherently locked into the Apple App Store with no alternative store options on this operating system. This is less true for apps on Google Play, where developers have flexibility to distribute their apps through third-party app stores that offers different in-app purchase policies. Therefore, the estimated effects of legislation on app performance within the Apple App Store ecosystem can serve as an upper bound, as substitution effects are limited.

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	1st %ile	25th %ile	Median	75th %ile	99th %ile
Downloads	16,705	56,533	6	452	2,341	11,319	254,605
Revenue	263,812	1,071,140	23	640	6,390	73,900	5,184,092
Observations	18,690						
Number of apps	174						
Number of developers	146						

Table 1 reports summary statistics for our 18,690 app-country-month observations. There is substantial variation in the size of apps – the 25th percentile app is installed 452 times per country per month and earns \$640 per country per month, while the 75th percentile app is installed over 11,000 times and earns over \$70,000 over the same period. Most developers (87%) release only one app, though we note that the developer information released by Appfigures may obscure corporate or conglomerate ownership structures.

Figure 1 and Figure 2 illustrate the distribution of app release years and primary app categories represented in the sample, respectively. Almost half of the apps that fit our criteria were released in 2023. While we can’t analyze the performance of these apps across the policy change directly, we touch on this phenomenon in the conclusion. Within our data the top categories were Games (Role Playing, Simulation, Strategy, and Action subcategories), followed by Social Networking apps.

4 Methods

To motivate our empirical work, we consider a static model in which the developer seeks to maximize the profit they earn from an app. Let p be the price of in-app purchases and w be the quality of the

Figure 1: Distribution of app release years

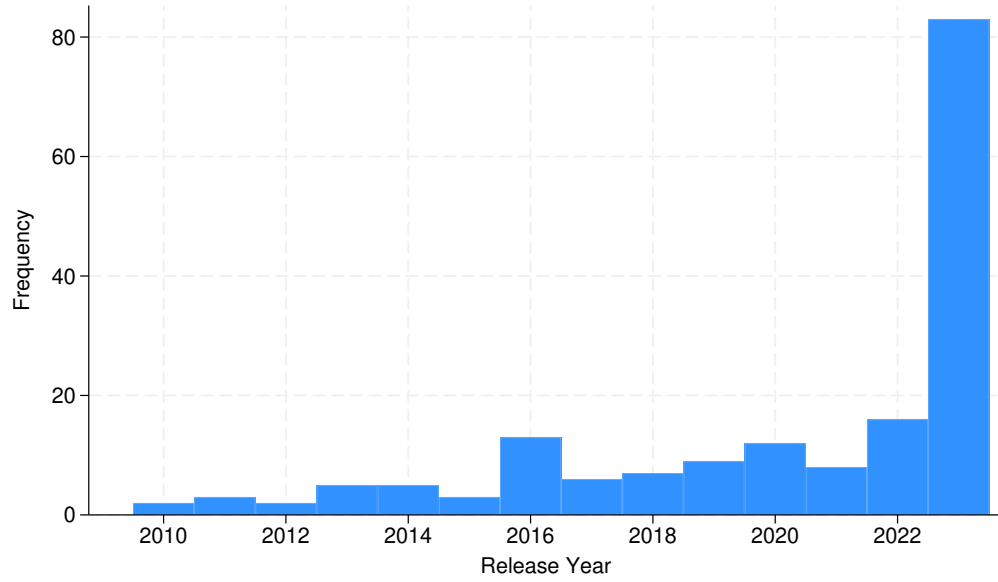
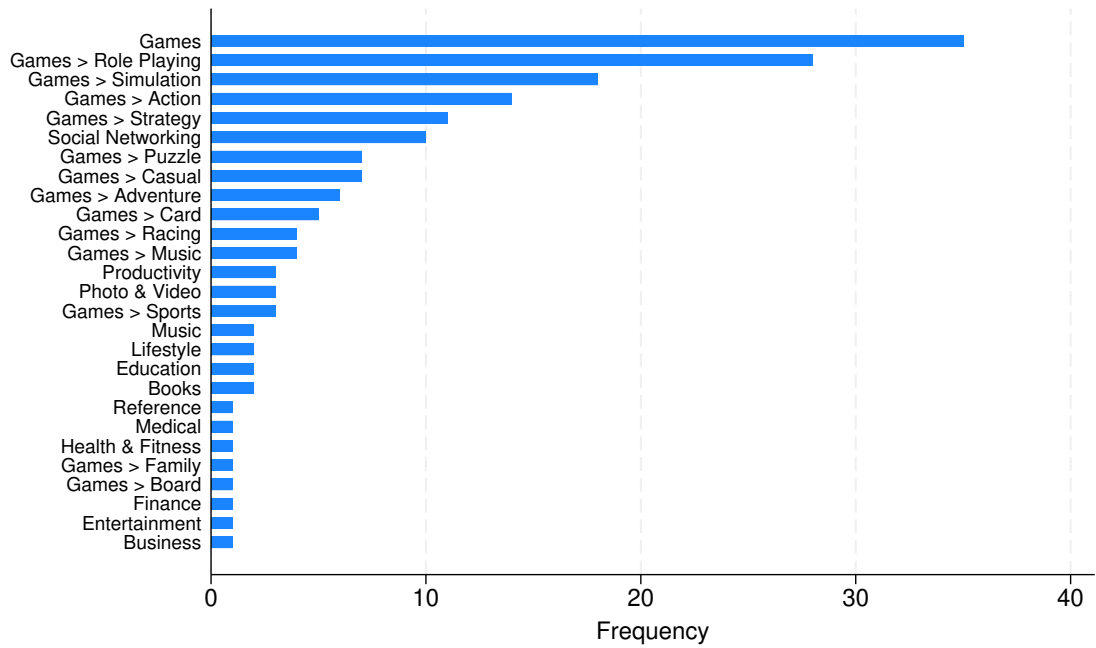


Figure 2: Distribution of app categories



app. Let $q(p, w)$ be the quantity of in-app purchases as a function of the in-app price and app quality, assumed to be increasing in w and decreasing in p . This encompasses both the quantity of downloads of an app and the average quantity of in-app purchases per user conditional on app download. Let the cost of development be $c(w)$ and let the marginal cost of providing purchases be $mc(w)$. This

encompasses, for example the costs involved in (a) developing in-app items or features to purchase and (b) tracking such purchases. Let the platform commission rate (as a fraction of revenue) be τ . The developer's problem is

$$\max_{p,w} (1 - \tau)q(p, w)[p - mc(w)] - c(w) \quad (1)$$

For simplicity in developing hypotheses, we specify tractable functional forms. Let $q(p, w) = Ap^{-\alpha}w^{\beta}$ and $c(w) = Bw^{\gamma}$, where α , β , and γ are strictly positive. Let $mc(w) = x$, a constant. By taking first-order conditions, it can be shown that the optimal price is $p^* = \frac{\alpha x}{\alpha - 1}$ and the optimal quality is

$$w^* = \left[(1 - \tau)(p^* - x)p^{*-\alpha} \frac{A\beta}{B\gamma} \right]^{\frac{1}{\gamma - \beta}}. \quad (2)$$

The derivative of w^* with respect to $(1 - \tau)$ is positive. Thus, to summarize, in this simple static framework, when the commission rate decreases, developers increase the quality of their apps and leave prices constant. We therefore form two hypotheses coming from the two components of the quantity of in-app purchases. A decrease in the commission rate should lead to

1. an increased number of downloads; and
2. an increase in total in-app revenue.

4.1 Empirical Approach

We estimate the effect of the policy change using a difference-in-differences model with fixed effects that compares the difference in outcomes for the treated group between the pre-treatment and post-treated periods to the difference in outcomes for the control group between the same periods.⁵ We estimate the parameters of

$$\log(Y_{ict}) = \beta_0 + \beta_1 D_{ict} + \alpha_i + \zeta_c + \gamma_t + u_{ict}, \quad (3)$$

where $\log(Y_{ict})$ is the logged outcome variable (either app installs or app revenue) for app i distributed in country c , at time t , D_{ict} is a treatment dummy that interacts PostReform_t , an indicator variable which is equal to 1 if t is June 2022 or later, and Treat_{ict} , an indicator variable which is

⁵Note that in this context, there is only one treated country, and so therefore many of the issues discussed by Callaway and Sant'Anna (2021) and Goodman-Bacon (2021) do not apply.

equal to 1 if app i for the users in country c is in the treated group. Fixed effects for cross sectional units are represented by α_i , ζ_c , which are app-level and country-level fixed effects, respectively. γ_t represents time fixed effects, and u_{ict} represents idiosyncratic app-country-time disturbances.

Although we have restricted our data to apps which are active (in the sense of strictly positive downloads and revenues) in the South Korean market, our data contain a few zeros. A visual inspection of the data reveals that many of these appear to be more accurately described as missing (for example, seeing a sudden drop in revenue to zero for one month and then a return to previous revenue levels), and so we drop those observations. We have also explored the possibility that these zeros are meaningful; there has been considerable debate about the appropriate estimation strategy to use in these cases (see e.g. Manning and Mullahy, 2001; Mullahy and Norton, 2022). As robustness checks, we use the Iterated Ordinary Least Squares method of Bellégo et al. (2022) and the two-part models of Belotti (2015) to re-estimate our parameters and find qualitatively similar results.⁶

The parameter of interest is β_1 , which represents the causal effect of Apple’s changed policy on the outcomes (app installs and app revenue), provided that the identification assumption is satisfied. For the difference-in-differences estimator to produce unbiased estimates, the parallel trend assumption must hold: the difference between the treated and the control remains constant throughout the analysis period in the absence of the treatment. Since dependent variable is logged, the estimate $\hat{\beta}_1$ can be interpreted as the percent change for treated apps during post-treatment period. $\hat{\beta}_1$ isolates the change in app installs and app revenue that cannot be explained by what was observed as the difference between the treated and control group in the pre-treatment period.

That said, it is possible that any results stemming from that control group are driven by seasonal trends in download and in-app purchase behavior. To account for this, we explore specifications using two sets of control groups in a triple-difference-style approach. That is, we estimate the parameters of

⁶Contact the corresponding author for details including regression tables.

$$\begin{aligned}
\log(Y_{ict}) = & \beta_0 + \beta_1 \text{PostYear}_t \times \text{PostMonth}_t \times \text{TreatCountry}_{ic} \\
& + \beta_2 \text{PostYear}_t \times \text{TreatCountry}_{ic} + \beta_3 \text{PostMonth}_t \times \text{TreatCountry}_{ic} \\
& + \beta_4 \text{PostYear}_t \times \text{PostMonth}_t \\
& + \beta_5 \text{PostYear}_t + \beta_6 \text{PostMonth}_t + \beta_7 \text{TreatCountry}_{ic} \\
& + \alpha_i + v_{ict},
\end{aligned} \tag{4}$$

where $\log(Y_{ict})$ is the logged outcome variable (either app installs or app revenue) at time (year-month) t for app i that belongs in country c , PostYear_t , and PostMonth_t are an indicator variable which is equal to 1 if t is 2022 or later, and July or later, and TreatCountry_{ic} is an indicator variable which is equal to 1 if the app i country c pair is treated. Fixed effects for cross sectional units (apps) are represented by α_i , and idiosyncratic disturbances are represented by v_{ict} .

4.2 Limitations

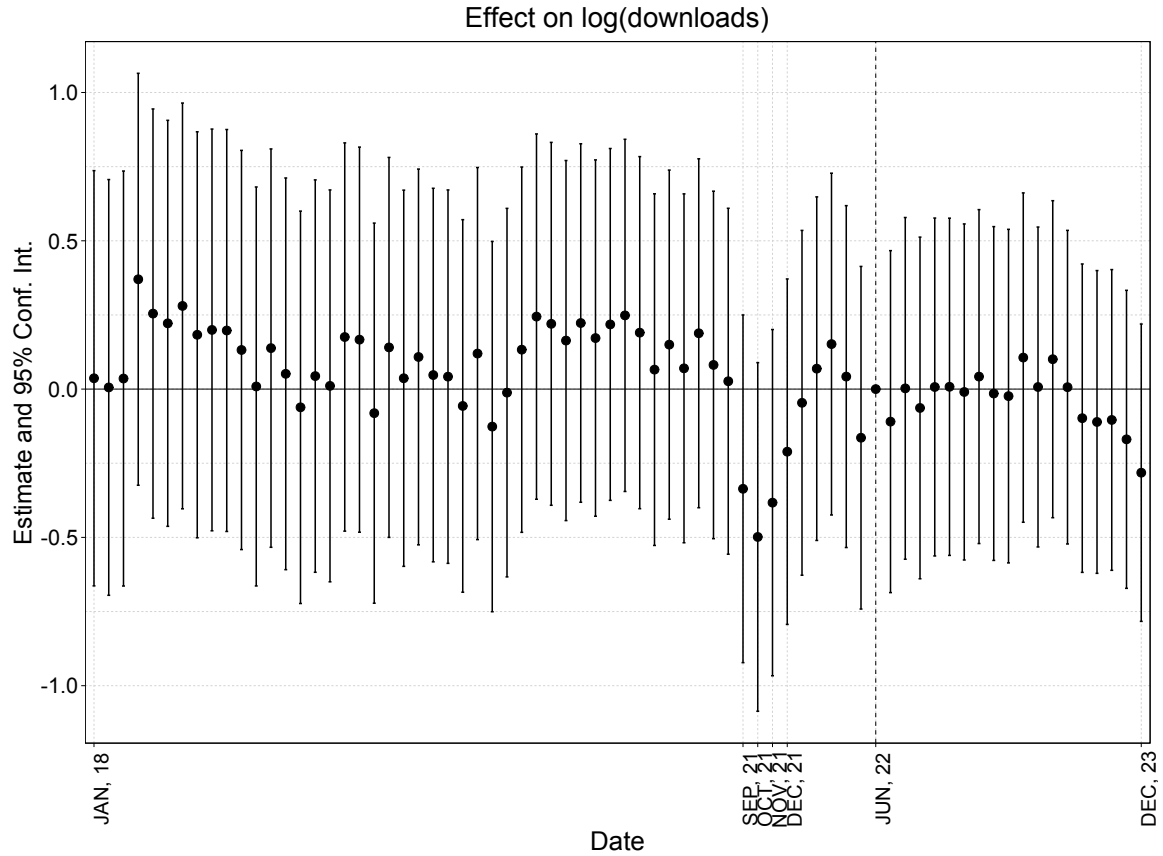
It is important to note that our analyses are subject to limitations. First, our identification strategies assume that no event (other than the policy change) systematically affected outcomes for apps distributed on the Apple App Store in South Korea during the analysis period. Second, given that the installed base of smartphone users in the United States is approximately an order of magnitude higher than in South Korea (and the number of smartphone users worldwide is another order of magnitude higher), it is reasonable to believe that firms serving global users and facing high development costs would forgo implementing alternative payment systems even in the context of South Korea’s law. Lastly, due to data limitations, we cannot directly observe app-level switching of billing systems, which left us to analyze indirectly the legislation’s effects through average app performance metrics, rather than directly counting affected apps and their individual performance changes caused by the legislation.

5 Results

5.1 Event Study

To ensure the ongoing consistency of app performance between Korea and other countries during the pre-treatment period, we conduct an event study. This methodology examines the parallel trend assumption in the pre-treatment period: accounting for app and country fixed effects, the average

Figure 3: Event Study of the Apple's Response on the Number of App Installs



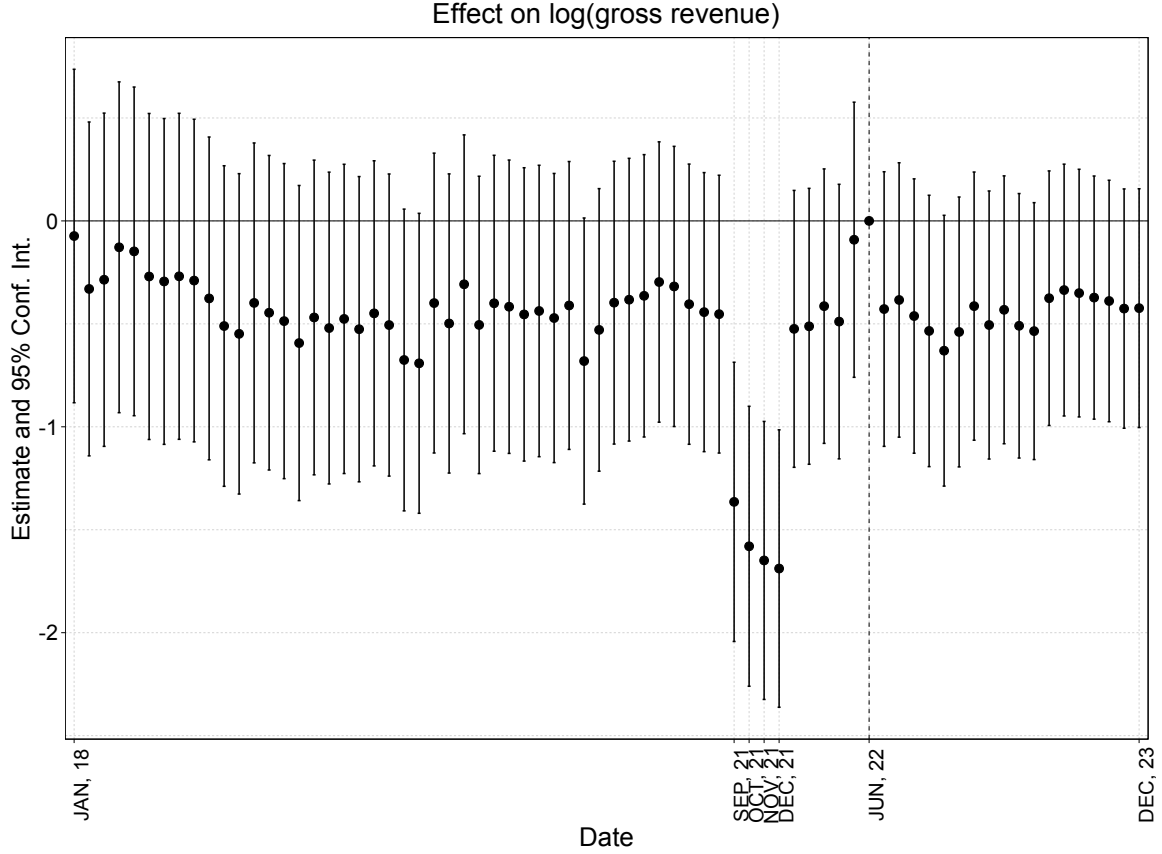
Notes: These estimates are from an event study regression. The dependent variable is logged number of downloads and the independent variables include app, country, and time fixed effects. The reference month is set on the month of June, 2022.

difference in app downloads during pre-treatment periods should not statistically differ from the period immediately preceding treatment onset.

Figure 3 illustrates the event study analysis of app downloads. We systematically compare app performance in the pre-treatment period against the period immediately preceding treatment initiation to identify any deviations in app behavior. The figure confirms no significant differences across time and countries before treatment, validating the parallel trend assumption. However, we do observe a potential drop from September 2021 through December 2021, though the confidence interval includes zero. This is a period of time of uncertainty in the market: the legislation had passed, but was not yet effective and neither Apple nor Google had not yet changed their policy.

Figure 4 presents the event study analysis of revenue. The estimates suggest there may be a consistent seasonal drop in revenue in South Korean (as compared to non-Korean markets) in November

Figure 4: Event Study of the Apple's Response on App Revenue



Notes: These estimates are from an event study regression. The dependent variable is logged app revenue and the independent variables include app, country, and time fixed effects. The reference month is June 2022.

and December of each year. As with the revenue data, we see a substantial drop in revenue for the fourth month period between September and December 2021. This could represent anticipation effects, or simply systematic bias in the data during those months as the effect of the legislation was not yet fully incorporated into Appfigures' collection methodology. To prevent any potential bias generated by this period, we exclude it from our analyses.

5.2 Using Apps in Countries with Closed Billing System as Control

We estimate Equation (3) using the `reghdfe` package (Correia, 2014) and cluster standard errors by app.⁷ Our estimation window consists of the six months prior to the reform through the six months after the reform (i.e. January 2022 through December 2022). Table 2 summarizes our estimates of β_1 . The point estimates suggest that downloads remain constant while revenue decreases slightly,

⁷Our primary concern is that there may be systematic variation in the accuracy of Appfigures' data by app.

Table 2: The effect of opening in-app billing systems using other countries as controls

	<i>Dependent variable:</i>	
	log(downloads)	log(revenue)
	(1)	(2)
Post * Treated	0.002 (0.064)	-0.093 (0.079)
Observations	3,302	3,302
R ²	0.765	0.826
RMSE	1.0677	1.295
F Statistic	0.00	1.38

Notes: The reported coefficients are estimates of β_1 in Equation (3), the interaction term between indicators for the post treatment period and the treated country (Korea). The dependent variables are the logged number of installs in Column (1) and logged revenue in Column (2). The specification includes fixed effects for app, country, and time. The window of analysis is January 2022 to December 2022. Standard errors are clustered at the app level and are reported in parentheses. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

though our estimates are noisy.

5.3 Using Triple Difference-in-Differences

We now turn to our estimates of Equation (4). We expand our analysis window to include January 2021 through December 2022 and once again cluster standard errors by app. The point estimates suggest the policy change may have led to an increase in downloads and, as before, a small decrease in gross revenue. Once again, however, our estimates are noisy and we can not make robust inferences.

5.4 Discussion

Taken together, a generous reading of the results suggest that the policy may have resulted in a slight increase in the number of installs and a decrease in the amount of revenue earned by apps on the Apple App Store in South Korea. The noise in the estimates, however, may lead someone to reasonably conclude that the the policy had no measurable effect. We discuss each of these possible interpretations in turn.

First, suppose that the causal effects were indeed in the direction estimated here: opening up

Table 3: Results using triple difference-in-differences

	<i>Dependent variable:</i>	
	log(downloads)	log(revenue)
	(1)	(2)
Treated Country * Treated Year * Treated Month	0.086 (0.104)	-0.027 (0.098)
Treated Country * Treated Year	-0.079 (0.126)	0.070 (0.137)
Treated Country * Treated Month	-0.102 (0.033)	-0.094 (0.059)
Treated Year * Treated Month	-0.118 (0.116)	-0.182** (0.076)
Constant	7.815*** (0.103)	9.645*** (0.110)
Observations	5,516	5,516
R ²	0.581	0.673
RMSE	1.465	1.752
F Statistic	2.17**	3.31***

Notes: The reported coefficients are estimates of β_1 in Equation (4), the interaction term between indicators for the post treatment year, post treatment months, and the treated country (Korea). The dependent variables are the logged number of installs in Column (1) and logged revenue in Column (2). The specification includes app fixed effects. The window of analysis is January 2021 to December 2022. Standard errors are clustered at the app level and are reported in parentheses. *p<0.1; **p<0.05; ***p<0.01.

in-app billing systems leads to increased downloads and decreased revenue. This result would correspond to one of our hypotheses. But why would revenue decrease? In the language of our model, and taking as given that downloads increase, the quantity of in-app purchases per consumer must decrease by more than the increase in downloads. What could explain that gap?

One potential answer lies in the trust that users have in third-party billing systems (Hillman and Neustaedter, 2017). Consumers may have less trust in third-party systems than in Apple’s billing system integrated to the store (and indeed the operating system itself), and this lack of trust may lead consumers to reduce the frequency with which they buy in-app items. Indeed, a study of third-party payment systems in other contexts revealed several potential security flaws (Yang et al., 2019). The simple need to choose a billing system may pose a sufficient friction to reduce the frequency of in-app purchases as well.

Perhaps more likely, however, is the null hypothesis: the Korean legislation and Apple’s policy change did not generate substantive changes in the mobile app market. This conclusion is eminently plausible both because Apple’s policy change consisted of an attempt to retain the vast majority of their commission rate, and because South Korea’s app market is small relative to the rest of the world. The incentives created by the policy change, while pushing in the direction of increased app quality (and therefore downloads and revenues) in South Korea, may not have been large enough to overcome the friction of development. Furthermore, for apps with a global reach, it is not necessarily clear how developers would focus on improving quality *for South Korea alone*.⁸

6 Conclusion

The near-monopoly power of the two leading app distribution platforms have raised antitrust concerns that these platforms leverage their market power to tie in-app billing system with their platform operations and charge developers a dictated commission rate that is higher than would be present in a more competitive market. To prevent leading platform operators from monopolizing in-app billing system, South Korea implemented an un-tying policy: app stores may not require developers to use any particular in-app billing system. This research examines the short-term effect of this legislation. We examine whether Apple App Store’s response to the legislation has any significant impact on app demand as well as app revenue. While the previous literature has largely focused on

⁸One notable possibility would be investing in improved language localization for apps with non-Korean design languages.

negative market consequence of tying (reducing incentives to innovate, price that is charged higher than competitive rate, etc.), the effects of unbundling are less well-studied. Our study is among the first to investigate the market consequence of decoupling in-app billing system from operation of app distribution platforms.

Though we focus our analysis on apps which are active in the South Korean market (i.e. that are most incentivized to respond to any policy change in that jurisdiction), we are unable to detect any substantial impact of the policy on observable outcomes including app installs and in-app gross revenue. We interpret this as informative: the impact of the policy change was *de minimis*. This is plausible in part because South Korea does not comprise a particularly large part of the app marketplace, in part because Apple's response was designed to limit the ability of developers to capture additional revenue from in-app purchases, and in part because the development and promotion cycle of apps may delay any response.

This leads us to caution against over-extrapolation from these results to other markets. If, for example, the United States or the European Union were to implement aggressive anti-tying and/or anti-competitive policies, one might reasonably expect the market to be more responsive. In these cases, our motivational framework would once again suggest that the quality of apps should increase, thus motivating additional downloads and in-app purchases from consumers.

We also note that other types of responses are possible. Our data contain a number of apps which entered in 2023, after the policy change went into effect. It is possible that this policy change may have ultimately encouraged developers to enter the market or to redesign apps monetized by an initial purchase price to feature in-app purchases instead. To the extent that policy makers in other jurisdictions contemplate similar (or more restrictive) anti-tying policies, they should consider the impacts of these policies on developers' design decisions, especially if there are policy preferences related to those design decisions (Berthon et al., 2019; Kleinman and Das, 2020). It remains to be seen how platform policies and marketplaces will evolve in response to the numerous attempts to restrict the market power of platform owners.

References

- B. Allyn. What the ruling in the Epic Games v. Apple lawsuit means for iPhone users. *NPR.org*, September 2021. URL <https://www.npr.org/2021/09/10/1036043886/apple-fortnite-epic-games-ruling-explained>.
- A. Amelio and B. Jullien. Tying and freebies in two-sided markets. *International Journal of Industrial Organization*, 30:436–446, 9 2012. ISSN 01677187. doi: 10.1016/J.IJINDORG.2012.03.002.
- P. Belleflamme and M. Peitz. Platform competition: Who benefits from multihoming? *International Journal of Industrial Organization*, 64:1–26, 2019.
- C. Bellégo, D. Benatia, and L. Pape. Dealing with logs and zeros in regression models. *arXiv preprint arXiv:2203.11820*, 2022.
- F. Belotti. twopm: Two-part models. *Stata Journal*, 15(1):3–20(18), 2015. URL <https://www.stata-journal.com/article.html?article=st0368>.
- P. Berthon, L. Pitt, and C. Campbell. Addictive de-vices: A public policy analysis of sources and solutions to digital addiction. *Journal of Public Policy & Marketing*, 38(4):451–468, 2019.
- B. Callaway and P. H. Sant’Anna. Difference-in-differences with multiple time periods. *Journal of econometrics*, 225(2):200–230, 2021.
- D. W. Carlton and M. Waldman. The strategic use of tying to preserve and create market power in evolving industries. *The RAND Journal of Economics*, 33:194–220, 2002. ISSN 07416261. doi: 10.2307/3087430.
- J. P. Choi and D. S. Jeon. A leverage theory of tying in two-sided markets with nonnegative price constraints. *American Economic Journal: Microeconomics*, 13:283–337, 2021. ISSN 19457685. doi: 10.1257/mic.20180234.
- J. P. Choi and C. Stefanadis. Tying, investment, and the dynamic leverage theory. *The RAND Journal of Economics*, 32:52–71, 2001. ISSN 07416261. doi: 10.2307/2696397.
- S. Correia. REGHDFE: Stata module to perform linear or instrumental-variable regression absorbing any number of high-dimensional fixed effects. Statistical Software Components, Boston Col-

- lege Department of Economics, July 2014. URL <https://ideas.repec.org/c/boc/bocode/s457874.html>.
- M. Coughlin, A. Ismail, and E. Keller. Apps with hardware: Enabling run-time architectural customization in smart phones. In *2016 USENIX Annual Technical Conference (USENIX ATC 16)*, pages 621–634, 2016.
- A. De Cornière, A. Mantovani, and S. Shekhar. Third-degree price discrimination in two-sided markets. 2023.
- F. Decarolis and M. Li. Regulating online search in the eu: From the android case to the digital markets act and digital services act. *International Journal of Industrial Organization*, 90:102983, 2023.
- T. Derdenger. Technological tying and the intensity of price competition: An empirical analysis of the video game industry. *Quantitative Marketing and Economics*, 12:127–165, 4 2014. ISSN 1573711X. doi: 10.1007/S11129-014-9143-9/TABLES/13. URL <https://link.springer.com/article/10.1007/s11129-014-9143-9>.
- A. Enache, R. Friberg, and M. Wiklander. Demand for in-app purchases in mobile apps—a difference-in-difference approach. *International Journal of Industrial Organization*, 88:102945, 2023.
- N. Eriksson, A. Gökhan, and M. Stenius. A qualitative study of consumer resistance to mobile payments for in-store purchases. *Procedia Computer Science*, 181:634–641, 2021.
- D. Ershov. Variety-based congestion in online markets: evidence from mobile apps. *American Economic Journal: Microeconomics*, 16(2):180–203, 2024.
- J. S. Gans. Remedies for tying in computer applications. *International Journal of Industrial Organization*, 29(5):505–512, 2011.
- J. S. Gans. Mobile application pricing. *Information Economics and Policy*, 24(1):52–59, 2012.
- R. Gomes and A. Mantovani. Regulating platform fees under price parity. *Journal of the European Economic Association*, page jvae014, 2024.
- A. Goodman-Bacon. Difference-in-differences with variation in treatment timing. *Journal of econometrics*, 225(2):254–277, 2021.

- J. Hamari, N. Hanner, and J. Koivisto. "why pay premium in freemium services?" a study on perceived value, continued use and purchase intentions in free-to-play games. *International Journal of Information Management*, 51:102040, 2020.
- S. Hillman and C. Neustaedter. Trust and mobile commerce in north america. *Computers in Human Behavior*, 70:10–21, 2017.
- S. Hwang and S. Kim. Google's app store policy change and the responses elicited from mobile content firms: A case study on korea's app store ecosystem. *Telematics and Informatics*, 72:101855, 2022. ISSN 0736-5853. doi: <https://doi.org/10.1016/j.tele.2022.101855>. URL <https://www.sciencedirect.com/science/article/pii/S0736585322000880>.
- S. Hyrynsalmi, A. Suominen, and M. Mäntymäki. The influence of developer multi-homing on competition between software ecosystems. *Journal of Systems and Software*, 111:119–127, 2016.
- R. A. Kleinman and S. Das. Excessive spending in a "free-to-play" smartphone game. *The American Journal on Addictions*, 29(6):528–530, 2020.
- V. Landsman and S. Stremersch. Multihoming in two-sided markets: An empirical inquiry in the video game console industry. *Journal of Marketing*, 75(6):39–54, 2011.
- W. G. Manning and J. Mullahy. Estimating log models: to transform or not to transform? *Journal of health economics*, 20(4):461–494, 2001.
- C.-H. Miao. The pricing of ancillary goods when selling on a platform. *International Journal of Industrial Organization*, 83:102847, 2022.
- J. Mullahy and E. C. Norton. Why transform y? a critical assessment of dependent-variable transformations in regression models for skewed and sometimes-zero outcomes. Technical report, National Bureau of Economic Research, 2022.
- J. Pybus and M. Coté. Super sdks: Tracking personal data and platform monopolies in the mobile. *Big Data & Society*, 11(1):20539517241231270, 2024.
- A. Ravoniarison and C. Benito. Mobile games: players' experiences with in-app purchases. *Journal of Research in Interactive Marketing*, 13(1):62–78, 2019.
- J.-C. Rochet and J. Tirole. Platform competition in two-sided markets. *Journal of the European*

- Economic Association*, 1:990–1029, 2003. doi: 10.1162/154247603322493212. URL <https://academic.oup.com/jeea/article/1/4/990/2280902>.
- I. Salehudin and F. Alpert. To pay or not to pay: understanding mobile game app users’ unwillingness to pay for in-app purchases. *Journal of Research in Interactive marketing*, 16(4):633–647, 2022.
- S. L. Schreft. Risks of identity theft: Can the market protect the payment system? *Economic Review-Federal Reserve Bank of Kansas City*, 92(4):5, 2007.
- D. Su, J. Liu, S. Zhu, X. Wang, W. Wang, and X. Zhang. Appq: Warm-starting app recommendation based on view graphs. *arXiv preprint arXiv:2109.03798*, 2021.
- T.-H. Teh, C. Liu, J. Wright, and J. Zhou. Multihoming and oligopolistic platform competition. *American Economic Journal: Microeconomics*, 15(4):68–113, 2023.
- Z. Wang and J. Wright. Should platforms be allowed to charge ad valorem fees? *The Journal of Industrial Economics*, 66(3):739–760, 2018.
- M. D. Whinston. Tying, foreclosure, and exclusion on jstor. *The American Economic Review*, 80: 837–859, 1990. URL <https://www.jstor.org/stable/2006711>.
- W. Yang, J. Li, Y. Zhang, and D. Gu. Security analysis of third-party in-app payment in mobile applications. *Journal of Information Security and Applications*, 48:102358, 2019.